

2-Year Software Engineering Career Roadmap

Personalized Strategic Plan — First-Year B.Tech CSE Student

Target: SDE Internship → Full-Time SDE Role

Current Position

1st-Year B.Tech CSE student with strong early foundation: C++, HTML, CSS, JS, React, Node.js, Git/GitHub, basic SQL & DSA. Ahead of peers in full-stack exposure.

Target Goal

Land a **product-based SDE internship** (Year 2–3), convert to a **full-time SWE role** by graduation. Build mastery in DSA, Full-Stack Dev, and applied AI tools.

Skill Gap Analysis

Area	Current Level	Target Level (for product-co internship)	Gap Priority
DSA & Problem Solving	Basic	500+ problems, contest-level pattern mastery	Critical
CS Fundamentals (OS, DBMS, CN, OOP)	Not started	Interview-ready depth	Critical
Full-Stack Development	Intermediate (MERN basics)	Production-grade projects, auth, deployment	High
SQL & Databases	Basic	Joins, indexing, schema design, NoSQL basics	High
System Design (Basic)	None	LLD + basic HLD for intern-level rounds	Medium
AI Tools & APIs	None	Use LLM APIs, prompt engineering, AI-assisted coding	Medium
Open Source / GitHub Presence	Basic Git use	Active contributions, strong profile	Medium

🌟 Recommended Learning Plan

Phase	Timeline	Focus Areas
Phase 1: Foundation	Months 1–6	DSA basics (arrays, strings, recursion, LL, stacks/queues) on C++; strengthen React/Node projects; SQL fundamentals; Git workflows
Phase 2: Depth	Months 7–12	Advanced DSA (trees, graphs, DP, greedy); OOP + DBMS + OS basics; build 2 full-stack projects; start CP (Codeforces/LeetCode)
Phase 3: Application	Months 13–18	System design basics (LLD); AI tool integration (OpenAI/Claude APIs); mock interviews; apply to internships
Phase 4: Placement Push	Months 19–24	Internship execution; polish resume/portfolio; advanced DSA revision; full-time SDE interview prep

Suggested Projects

- Full-Stack SaaS App** — MERN + auth + payments + deployment (Vercel/Render)
- AI-Powered Tool** — React app using an LLM API (e.g., resume analyzer, chatbot)
- DSA Visualizer** — Interactive algorithm/data structure visualizer (JS/React)
- Open Source Contribution** — 3–5 merged PRs to active GitHub repos

Networking Strategy

- Optimize LinkedIn: headline, banner, weekly build-in-public posts
- Join college DSA/dev clubs; participate in hackathons (2–3/year)
- Follow & engage with SDEs at target product companies
- Attend tech meetups / Discord communities (DSA, webdev, AI)
- Cold-message alumni at target companies for referrals
- Maintain active GitHub with pinned, well-documented repos

Monthly Milestones

Timeframe	Milestone
Month 3	150+ DSA problems solved; 1 polished full-stack project deployed
Month 6	SQL + Git mastery; LinkedIn & GitHub profiles optimized; first hackathon
Month 9	300+ DSA problems; DBMS/OOP fundamentals solid; 2nd major project live
Month 12	500+ DSA problems; 1st open-source PR merged; resume v1 ready
Month 15	LLD basics done; AI-integrated project built; start internship applications
Month 18	Mock interviews (10+); internship offer secured
Month 21	Internship in progress; strong recommendation/network built
Month 24	Full-time SDE offer(s) via internship conversion or campus placement

Immediate Next Actions (This Week)

- Set up LeetCode/Codeforces profile; solve 5 easy DSA problems today
- Create/optimize LinkedIn + GitHub profiles with clear headline & bio
- Pick and scope your first full-stack project idea
- Join 1 college coding club or Discord DSA community
- Block a fixed daily 1-hour DSA + 1-hour build slot in your calendar